



Dual disentanglement of user–item interaction for recommendation with causal embedding

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ABSTRACT

To achieve personalized recommendations, the recommender system selects the items that users may like by learning the collected user–item interaction data. However, the acquisition and use of data usually form a feedback loop, which leads to recommender systems suffering from popularity bias. To solve this problem, we propose a novel dual disentanglement of user–item interaction for recommendation with causal embedding (DDCE). Different from the existing work, our innovation is we take into account double-end popularity bias from the user-side and the item-side. Firstly, we perform a causal analysis of the reasons for user–item interaction and obtain the causal embedding representation of each part according to the analysis results. Secondly, on the item-side, we consider the influence of item attributes on popularity to improve the reliability of the item popularity. Then, on the user-side, we consider the effect of the time series when obtaining users' interest. We model the contrastive learning task to disentangle users' long–short-term interests, which avoids the bias of long–short-term interests overlapping, and use the attention mechanism to realize the dynamic integration of users' long–short-term interests. Finally, we realize the disentanglement of user–item interaction reasons by decoupling user interest and item popularity. We experiment on two real-world datasets (Douban Movie and KuaiRec) to verify the significance of DDCE, the average improvement of DDCE in three evaluation metrics (NDCG, HR, and Recall) compared to the state-of-the-art model are 5.1106% and 4.1277% (MF as the backbone), 3.8256% and 3.2790% (LightGCN as the backbone), respectively.

1. Introduction

The purpose of a recommender system is to supply an individuation service for the user. At present, it gives scope to a significant role in applications. For instance, video streaming media (Liu, Xu, et al., 2023), online shopping (Zhan & Xu, 2023), and social media. Data acquisition and use usually form a feedback loop in the recommender system. The recommender system trains the collected user–item interaction data and generates recommendations for users. Then users interact with the item according to the recommendation results to generate user feedback data. Through a certain number of iteration cycles, the recommender system is suffered from a popularity bias problem easily. Take the movie recommendation as an example, demonstrated in Fig. 1. As can be observed from the figure, when the user watches comedy movie 1 and gives positive feedback, the recommender system trains the obtained data and then continues to recommend the comedy movie to the user. Comedy movies will gain higher weight in training, and users may be satisfied with the recommended results during a short period. But with the passage of time, the recommender

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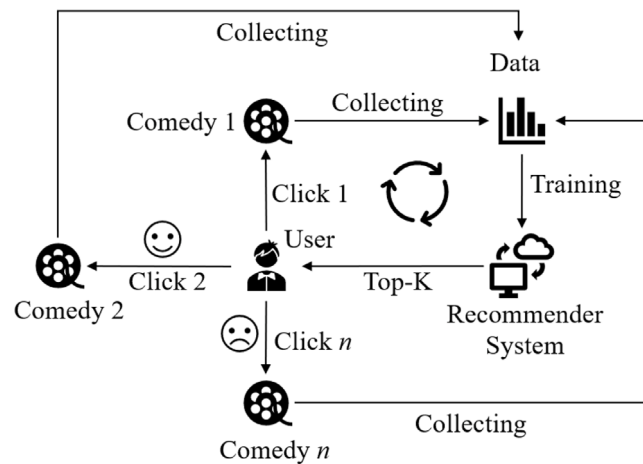


Fig. 1. The feedback loop of user-item interaction and recommender system.

system will converge to recommend comedy movies to users by degrees, which will make users fall into the filtering bubble and eventually lead to users dissatisfaction. For this reason, it is important to take into account the popularity bias in the individuation recommendation.

There are generally four methods to solve the popularity bias problem: (1) Inverse propensity scoring (IPS). The uniformity of data distribution is adjusted by adjusting the weights of interactive examples trained by the model (Gruson et al., 2019; Schnabel et al., 2016) so that items with higher popularity have lower weights and those in the long tail have higher weights. Although IPS has a good theoretical basis, the variance of the model is large. (2) Ranking adjustment, reordering the recommendation list (Abdollahpouri et al., 2019; Zhu et al., 2021) to raise the score of the not popular item. (3) Regularization training (Bonner & Vasile, 2018), extracting information from the unified data to solve the popularity bias. Ranking adjustment and regularization training both depend on the heuristic design of researchers, and the accuracy of recommendations is usually affected. (4) Causal embedding (Liu et al., 2020; Wei et al., 2021) separates popularity bias from user preferences through causal reasoning.

Recently, with the advances in research on causal reasoning, the application of the recommender system in causal reasoning makes recommendations more explanatory and stable. When disentangling popularity bias from user preferences, most existing causal recommendation methods usually eliminate the popularity bias directly. However, the popularity bias does not mean a bad effect, and the elimination without consideration will lead to the loss of some information. Therefore, some studies have made use of the popularity bias, such as distinguishing useful popularity bias from harmful popularity bias (Zhao, Chen, et al., 2022) or adjusting the recommended score to the expected popularity bias through intervention (Zhang et al., 2021). Through the above research, we found that the current model of debiased recommendation based on causal embedding still faces the following challenges:

- (1) The popularity bias in these methods is usually handled on the user or the item-side (Yalcin & Bilge, 2022), but each of these biases can affect the effectiveness of recommendations. Therefore, the first challenge is how to consider the influence of popularity bias on recommendations on both sides.
- (2) When distinguishing user interest and item popularity, user interests are usually treated as stable. However, in the real-world recommender system, the user's interest is usually composed of two parts: steady long-term interests and short-term interests influenced by time. So, how to reasonably utilize these two interests in the decoupling process is the second challenge.
- (3) The reason users interact with the item may be caused by the conformity of the popular item. When disentangling user interest and item popularity, the item popularity is usually directly expressed by conformity, neglecting the influence of the attributes of the item on popularity, which is unfair to recommend items with high quality but low conformity. Therefore, the third challenge in the study of item popularity is how to use the attributes of the item to improve recommendation performance.

The objective of our research is to analyze the causality of user-item interactions, and to achieve the dual disentanglement of the user-item interaction to address the above challenges by researching the popularity bias from both the user and the item-side based on the causal embedding. In this paper, we present a fresh dual disentanglement recommendation model of user-item interaction with causal embedding. We first conduct a causal analysis of user-item interaction and divide the reasons for interaction into user interest and item popularity. On the basis of causality, we use separate causal embedding for each part of the user and the item. Then, based on causal embedding, the model takes into account the user and the item-side popularity bias when disentangling user interests and item popularity. On the item-side, we set the popularity of the item as its attributes and conformity to enhance the dependability of item popularity. On the user-side, we divide the user interest into long-term interest and short-term interest. For fear of the entanglement of these two interests, we propose a contrastive learning task to disentangle users' long-short-term interests. At the same time, we propose a dynamic integration method of long-short-term interests to enhance the precision and

interpretability of the model. Finally, the user–item interaction is disentangled on the basis of the causal analysis to realize the dual disentanglement of user–item interaction with causal embedding and reduce the effect of popularity bias on the recommendation. The main work and contributions of this paper are indicated below:

- (1) We consider the popularity bias from a double-end perspective in the recommendation. The causal embedding of each part of the interaction between users and items is obtained based on the causality. And design the contrastive learning task to achieve dual disentanglement of the user–item interaction so that the model has better interpretability.
- (2) We take into account the item attributes in the item’s popularity to make the adverse impact of conformity lessen on recommendations.
- (3) When acquiring user interests, we model long–short-term interests separately according to causal embedding and disentangle them. For different users and items, the consequence of users’ long–short-term interests in interaction is different. We propose a dynamic integration of users’ long–short-term interests, which enables the model better adapt to different scenarios and enhances its robustness of the model.
- (4) We adopt two real-world datasets to compare DDCE with benchmarks, and the experimental findings validate the efficiency of the DDCE. We conducted ablation experiments to study how various parts of DDCE affect recommendations.

The theoretical and practical implications of our research are that we consider the influence of popularity bias on recommendation from a double-end perspective when studying popularity bias based on causal embedding, which makes the study of popularity bias more comprehensive. We consider item attributes in the research of item popularity which is helpful to alleviate the problem of item-side popularity bias. We research the disentanglement and dynamic integration of users’ long–short-term interests, which makes our recommendation model closer to the real-world recommender system.

In this paper, Section 2 discusses research on popularity bias in the recommendation and causal recommendation. Section 3 introduces the dual disentanglement of user–item interaction for recommendation with causal embedding. Section 4 describes the process of experimental and analyzes experimental results. Section 5 summarizes the work of this paper and considers the next research.

2. Related works

In this section, we first introduce the related works on popularity bias in recommendation, then introduce the related works on causal recommendation, and finally introduce the current research progress on popularity bias based on causal recommendation.

2.1. Popularity bias in recommendation

Popularity bias often makes popular users or items gain higher weight in model training, which can have negative effects on recommendations, such as trapping users in filter bubbles and the recommendation results are unfair (Zhu et al., 2021). Therefore, the issue of popularity bias in recommender systems needs to be addressed urgently. Research on popularity bias can help alleviate unfairness in recommendations (Abdollahpouri et al., 2020; Li et al., 2021; Morik et al., 2020; Yalcin & Bilge, 2023; Zhang et al., 2021). IPS is widely used in the study of popularity bias. For example, Ai et al. (2018) proposed the DLA model. This is an automatic unbiased learning ranking framework, learning an unbiased ranking model directly from biased click data. Zhu et al. (2020) studied the generation of unbiased recommendations according to the biased implicit interaction of user–item. They proposed a framework of combinatorial joint learning that learns unbiased user–item correlation and unbiased orientation at the same time. However, the variance of the inverse propensity scoring model is usually large (Xiao & Wang, 2022), so some researchers use ranking adjustment and regularization training methods. For example, Yalcin and Bilge (2021) studied the issue of popularity bias in user group recommendations and used a reordering method to neutralize potential popularity bias issues in group recommendations. Abdollahpouri et al. (2021) calibrated the recommendation list to users based on their tolerance for popular items. Bonner and Vasile (2018) proposed learning from large samples of biased and unbiased exposure feedback data to decompose the user implicit feedback matrix to optimize recommendation results. Zhu et al. (2021) regularized the popularity of items and studied the popularity bias problem from the perspective of equal opportunity. Xv et al. (2022) proposed the PID model that uses two strategies to neutralize the popularity bias by adding regularization terms to the loss function. Chen et al. (2022) proposed the CD²AN method, which separates item attributes and popularity through regularization to learn biased and unbiased models. The methods of ranking adjustment and regularization training rely on the heuristic design of researchers, and the accuracy of recommendations is often affected. Therefore, some researchers have studied popularity bias from the perspective of causal reasoning. We will provide a detailed introduction to the study of popularity bias based on causal recommendations in Section 2.3.

2.2. Causal recommendation

For the past few years, with the research and development of causal reasoning, it has been used to solve the problems in recommender systems. For instance, Zhai et al. (2023) proposed a causality CTR prediction based on online advertising, considering the causality between the features in the graph. Tan et al. (2021) proposed the CountER model to obtain an understanding of counterfactual reasoning from causal reasoning for interpretable recommendations. Subsequently, from the perspective of causal reasoning, they proposed to introduce counterfactual and factual reasoning to resolve the issue of interpretable GNN (Tan et al.,

Table 1
Summary of work on popularity bias based on causal recommendation.

	DICE	PDA	MACR	CauSeR	CaDSI	TIDE	LAiR
user-side	✓				✓		✓
item-side		✓	✓	✓		✓	✓

(2022) with learning and evaluation problems. According to the inverse propensity scoring technology, Sato et al. (2020) proposed a framework with unbiased learning for the causal effect of recommendation, which is used to expand the causal effect of ranking indicators. Li et al. (2022) implemented causal feature selection in FMs to improve recommendations stability when there is a difference in the distribution of training data and test data. Inspired by the hybrid analysis in causal reasoning, Li et al. (2023) proposed an unbiased robust method from the causality perspective, DENC, to study missing-not-at-random (MNAR) problems in recommender systems. Gao, Lei, et al. (2022) proposed the CIRS model to use the counterfactual interactive in the recommender system. The model used causal reasoning to enhance offline reinforcement learning and handled the filtering bubble issue in the interactive recommendation. Du et al. (2021) proposed ExCAR, an interpretable causal reasoning framework for knowledge enhancement of event graphs. The model obtains additional evidential information from causal event graphs as logical rules of causal reasoning. Xu et al. (2022) proposed the ∂ CCF model to estimate user preferences for items after the intervention. This is a dynamic causal collaborative filtering method based on backdoor adjustments and alleviates the echo chamber through counterfactual reasoning. Xiao and Wang (2022) studied the optimization problem of ranking metrics for unbiased and robust causal estimation based on causal recommender systems. Wang, Feng, et al. (2022) proposed a cause-enhanced user-controllable inference framework UCI that enables users to actively control filter bubbles. Wei et al. (2022) proposed a causal recommendation model KGCR based on the knowledge graph, which uses counterfactual reasoning to eliminate the bias in the user-item similarity score. Some researchers also study unfairness in recommender systems from the perspective of causality. For example, Wu et al. (2019) constructed a structural causal model to study the problem of unrecognizable counterfactual quantities in counterfactual fairness. Zhang et al. (2020) used a multi-task learning framework to handle post-click conversion rate (CVR) estimation in industrial environments from a causal perspective. Li et al. (2021) studied the counterfactual fairness of user personalization in recommender systems based on the concept of causal fairness. Huang et al. (2022) combined causal reasoning with the bandit model and proposed a fair causal robber model (F-UCB) to realize the fairness of counterfactual individuals in the recommendation.

2.3. Popularity bias based on causal recommendation

The application of causal reasoning in the recommender system has demonstrated its good performance. With the development of causal recommendation, some researchers have used causal reasoning to solve the challenge of popularity bias in the recommender system. Such as Zheng et al. (2021) proposed a DICE framework using causal embedding to solve the entanglement of user interest and conformity in the recommender system. Wei et al. (2021) proposed the MACR model, which described the causality among variables in the recommender system from the causal inference perspective and eliminated the influence of item popularity on the recommendation by counterfactual reasoning. Some researchers study and apply popularity bias rather than eliminate it. Zhang et al. (2021) researched the influence of popularity bias by causal graphs in the recommender system and proposed a popularity bias disambiguation and adjustment framework PDA, which adjusted the recommendation scores to expected according to the causal intervention. Gupta et al. (2021) proposed the CauSeR model and studied the popularity bias in the session-based recommender system. They considered the popularity bias in the data generation and model training period. Wang, Li, et al. (2022) proposed a causal disentanglement model called CaDSI for semantic perceptual intention learning, which intervenes in user representation through causality to remove mixed biases caused by heterogeneous information. Zhao, Chen, et al. (2022) proposed a time-aware disentanglement framework (TIDE), which uses time information for disentanglement training so that recommendations can benefit from useful popularity bias and avoid harmful popularity bias. Liu, Fang, and Wu (2023) proposed LAiR, which is a fair-centered model alleviates that the local popularity bias of each user and item through a fair discriminator. We summarized the above research work on popularity bias based on the causal recommendation from the perspective of the user and the item, as shown in Table 1, where “✓” indicates that this work was studied from this perspective.

Although the above research on solving the popularity bias in the recommender system through causal reasoning has made some progress, we found from the summary in Table 1 that they usually address the popularity bias of the user or the item-side, neglecting the common influence of both. LAiR has conducted research from both the user and the item-side, but it is centered around fairness. Different users interact with different items for different reasons, so the bias from the model is different on the user and the item-side. Consequently, it is significant to think about the impact of both sides when dealing with popularity bias. In this research, we propose a model to disentangle the user-item interaction that can consider both the popularity bias of the user and the item-side, making the recommendation more explanatory. Meanwhile, we analyze user interests and item popularity when using causal reasoning to disentangle user-item interactions. Our model divides user interests into long-term interests and short-term interests and performs disentanglement and dynamic integration on them. For the popularity of the item, our model considers the attributes of the item, which alleviated the negative influence of conformity on interaction.

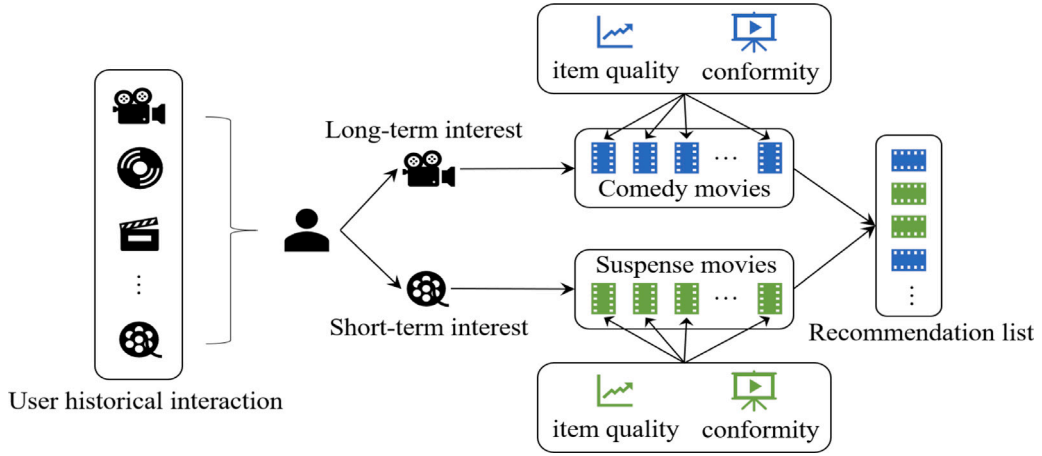


Fig. 2. An example of the movie recommendation.

Table 2
Key symbols used in this study.

Notation	Description
y_{uv}	The user-item interaction scoring
y_{int}	The user interest scoring
y_{pop}	The item popularity scoring
e^u, e^v	The embedding of users and items
$\mathcal{H}_u$	The historical interaction behavior sequence of user u
L_u^l	Users' long-term interests
S_u^s	Users' short-term interests
L_u^t	The contrastive representation of users' long-term interests
S_u^t	The contrastive representation of users' short-term interests
u_{int}	The interest of user u
p_v	The popularity of item v
q_v	The attribution of item v
c_v	The conformity of item v

3. Dual disentanglement of user-item interaction for recommendation with causal embedding

Our model consists of the following three stages: causal embedding, acquisition of users' interests, and user-item interaction disentanglement. In the causal embedding stage, we first define a causal graph of user-item interaction and set causal embedding based on the causal graph. On the item-side, we consider the influence of item attributes on item popularity to improve the reliability of the popularity. In the acquisition of users' interests stage, we disentangle and dynamically integrate long-short-term interests, enabling our model to adapt to different scenarios better. In the user-item interaction disentanglement stage, we disentangle user-item interaction based on causality to make the popularity bias adverse impact decrease in the recommendation. To further understand the model proposed in this paper, we use the movie recommendation as an example to explain our model, as shown in Fig. 2.

Firstly, by analyzing the historical and recent data of users watching movies, we found that users have a long-term preference for watching comedy movies and a recent preference for watching suspense movies. Then, the obtained long-term and short-term interests of users are dynamically integrated, and when recommending movies to users, both comedy movies and suspense movies are recommended. Finally, taking into account the score (item quality) of the recommended movie and the total number of times users have watched this movie (conformity), the final recommendation list is generated for the user. The key symbols used in this study are shown in Table 2.

3.1. Causal embedding

The user-item interaction depends on two aspects: user interest and item popularity. We construct a user-item interaction causal graph to represent the interaction process of the user and the item and the causality between variables according to these two aspects, demonstrated in Fig. 3.

According to the user-item interaction cause graph, we describe the score of user-item interaction as Eq. (1).

$$y_{uv} = y_{int} + y_{pop} \quad (1)$$

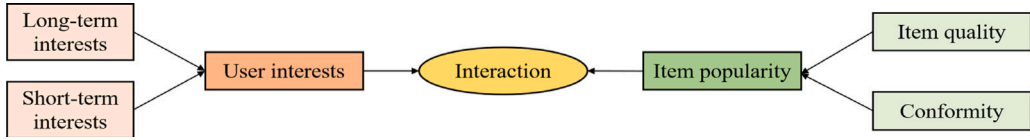


Fig. 3. User-item interaction cause graph.

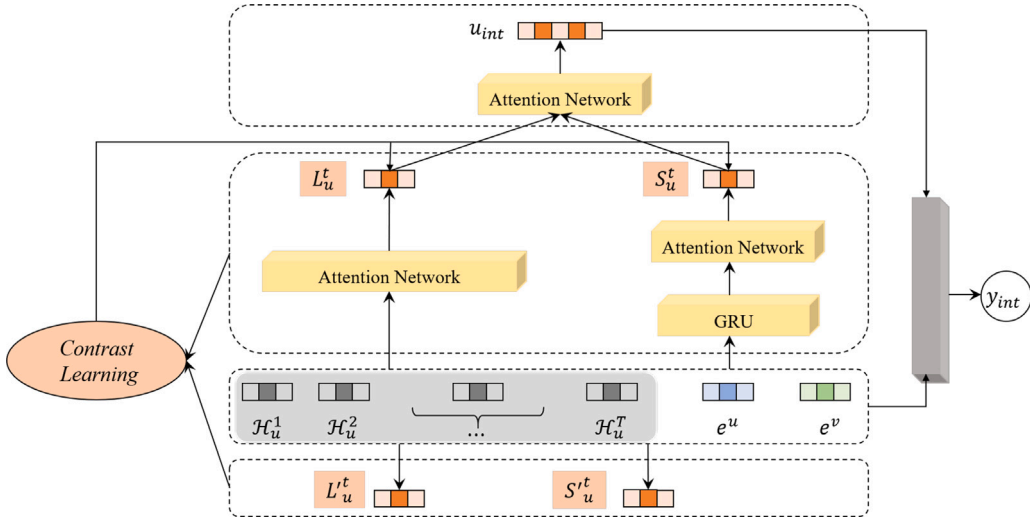


Fig. 4. The framework for the acquisition of users' interests.

where y_{uv} represents the overall interaction score, y_{int} represents the user interest score, and y_{pop} represents the item popularity score. The user's interaction with the item may depend on one or both of these reasons. Therefore, this paper conducts causal modeling for these two causes separately, rather than using the same method to express them uniformly.

According to the cause graph, we set up the following causal embedding. We use $\mathcal{U}$ to represent the user set, $\mathcal{V}$ to represent the item set, $\mathcal{U} = \{u_1, u_2, \dots, u_M\}$, $\mathcal{V} = \{v_1, v_2, \dots, v_N\}$, where M and N represent the number of users and items, respectively. The embeddings of users and items are represented as $e^u \in \mathbb{R}^{k \times M}$, $e^v \in \mathbb{R}^{k \times N}$, respectively, where k represents the embedding size. On the user-side, user interests consist of long-term interests and short-term interests. The interaction items of user u are sequentially collected before time T to obtain the historical interaction behavior sequence $\mathcal{H}_u$ of user u , $\mathcal{H}_u = [\mathcal{H}_u^1, \mathcal{H}_u^2, \dots, \mathcal{H}_u^T]$, where $u \in \mathcal{U}$, $\mathcal{H}_u^t \in \mathcal{V}$. The users' long-short-term interests are expressed according to the user-item historical interaction. On the item-side, we divide item popularity into attributes and conformity. We use q_v to represent the causal embedding of item attributes, c_v to represent the causal embedding of conformity. For example, in Douban movies, users can comment on movies they have watched. The attributes q_v of the movie are determined by the average score of all users who have watched the movie. The score is relatively stable, and the higher the comprehensive score, the higher the quality of the movie. The conformity c_v of the movie is usually expressed as the number of interactions observed on movie v , which is easily affected by time. The popularity of items is obtained by inner product $p_v = \langle q_v, c_v \rangle$. In the meantime, to project the item popularity to the $[0, 1]$ interval by the tanh nonlinear activation function to obtain the item popularity score, expressed as Eq. (2).

$$y_{pop} = \tanh(\langle q_v, c_v \rangle) \quad (2)$$

3.2. Acquisition of users' interests

User interests consist of relatively stable long-term interests and dynamic short-term interests. In the existing research on users' long-short-term interests, the acquisition of long-term interests usually adopts the collaborative filtering model such as matrix decomposition, while the acquisition of short-term interests adopts the sequential model. The long-short-term interests may be entangled together. Therefore, when acquiring user interests from user-item historical interaction data, this paper models the two interests based on causal embedding separately, rather than using a unified representation. The framework for the acquisition of users' interests is demonstrated in Fig. 4.

The acquisition of user interest is divided into the following three aspects: (1) Learning the users' long-term and short-term interests separately from the user-item interaction sequence based on causal embedding. (2) Calculating the similarity between users' long-short-term interests and their contrastive representation, designing a contrastive learning method to compare them, and

disentangling users' long-short-term interests. (3) Dynamic integration of users' long-short-term interests to obtain users' interests, and then calculate users' interest scores.

3.2.1. User long-term interest acquisition

Users' long-term interests reflect their relatively stable preferences, which are less influenced by time. Therefore, users' long-term interests can be obtained according to their historical interaction information. In user historical interaction sequences, different items have different effects on user interests. Therefore, to acquire the attention score for each item, this paper uses the attention mechanism, as formulated in Eqs. (3) and (4).

$$v_l = \sigma(W_1 \cdot e^v) \quad (3)$$

$$\alpha_v = MLP[v_l \| e^u \| (v_l \cdot e^u)] \quad (4)$$

The embedding v_l of the item during users' long-term interests is obtained by transformation matrix W_1 . Where σ represents the nonlinear activation function $\tanh$, $\|$ represents the connection between embeddings. We use multilayer perceptron (MLP) to connect embeddings and use the Softmax function to normalize it, as formulated in Eq. (5).

$$\tilde{\alpha}_v = \frac{\exp(\alpha_v)}{\sum_{v=1}^t \exp(\alpha_v)} \quad (5)$$

Users' long-term interests L_u^t can be obtained by weighting and aggregating the entire historical interaction information. Based on the above item attention score, it can be calculated as Eq. (6).

$$L_u^t = \sum_{v=1}^t \tilde{\alpha}_v \cdot e^v \quad (6)$$

3.2.2. User short-term interest acquisition

The sequence of user interactions influences on users' short-term interests. As users' interactions with the item change over time, their short-term interests also change. Recurrent neural networks have a good effect on data mining with sequential characteristics, so we input user history interaction items into the RNN model. The Gate Recurrent Unit (GRU) is the RNN model used in this study. Then, we acquire the attention scores using the attention mechanism to obtain the attention scores of the item during short-term interest. Finally, MLP is used to splice the embeddings, as given in Eqs. (7)–(9).

$$[e_1^v, e_2^v, \dots, e_t^v] = GRU[H_u^1, H_u^2, \dots, H_u^T] \quad (7)$$

$$v_s = \sigma(W_2 \cdot e_s^v) \quad (8)$$

$$\beta_v = MLP[v_s \| e^u \| (v_s \cdot e^u)] \quad (9)$$

The embedding of the item during users' short-term interests is obtained by transformation matrix W_2 , where $e_s^v \in [e_1^v, e_2^v, \dots, e_t^v]$. Then, we use the *Softmax* function to normalize it, as formulated in Eq. (10).

$$\tilde{\beta}_v = \frac{\exp(\beta_v)}{\sum_{v=1}^t \exp(\beta_v)} \quad (10)$$

According to the calculated attention score of the item in short-term interest, the short-term interest of users can be calculated as Eq. (11).

$$S_u^t = \sum_{v=1}^t \tilde{\beta}_v \cdot e_s^v \quad (11)$$

3.2.3. User long-short-term interests disentanglement

In this paper, separate modeling of user interest acquisition is realized in Sections 3.2.1 and 3.2.2. However, due to the unsupervised method used for interest acquisition, the disentanglement of users' long-short-term interests cannot be guaranteed. To address this issue, inspired by literature (Zheng et al., 2022), we design a contrastive task to disentangle users' long-short-term interests. We use the whole historical interaction behavior to obtain the long-term interest contrastive representation of users. And obtain short-term interest contrastive representation through the average value of k items that users have interacted with recently. Therefore, the formula of contrastive representation for the long-short-term interests of user u at time t as given in Eqs. (12) and (13).

$$I_u'' = \frac{1}{t} \sum_{v=1}^t e^v \quad (12)$$

$$S_u'' = \frac{1}{k} \sum_{v=1}^k e^{t-v+1} \quad (13)$$

Then calculate the similarity between users' long-short-term interests and the contrastive representation of users' long-short-term interests to realize the disentanglement of users' long-short-term interests. Contrastive learning is shown in Eqs. (14)–(17).

$$\text{Sim}(L_u^t, L_u^t) > \text{Sim}(L_u^t, S_u^t) \quad (14)$$

$$\text{Sim}(L_u^t, L_u^t) > \text{Sim}(L_u^t, S_u^t) \quad (15)$$

$$\text{Sim}(S_u^t, S_u^t) > \text{Sim}(S_u^t, L_u^t) \quad (16)$$

$$\text{Sim}(S_u^t, S_u^t) > \text{Sim}(S_u^t, L_u^t) \quad (17)$$

Finally, we use the paired loss function BPR (Bayesian Personalized Ranking) to complete the above four contrastive tasks, making the user's long-short-term interests closer to their corresponding positive samples, as formulated in Eqs. (18)–(21).

$$\mathcal{L}_1 = \rho(\langle L_u^t, S_u^t \rangle - \langle L_u^t, L_u^t \rangle) \quad (18)$$

$$\mathcal{L}_2 = \rho(\langle L_u^t, S_u^t \rangle - \langle L_u^t, L_u^t \rangle) \quad (19)$$

$$\mathcal{L}_3 = \rho(\langle S_u^t, L_u^t \rangle - \langle S_u^t, S_u^t \rangle) \quad (20)$$

$$\mathcal{L}_4 = \rho(\langle S_u^t, L_u^t \rangle - \langle S_u^t, S_u^t \rangle) \quad (21)$$

where ρ represents the nonlinear activation function ReLU , $\langle \cdot, \cdot \rangle$ represents the inner product between two embeddings. The disentanglement contrastive loss function of users' long-short-term interests is described as Eq. (22).

$$\mathcal{L}_{int} = \mathcal{L}_1 + \mathcal{L}_2 + \mathcal{L}_3 + \mathcal{L}_4 \quad (22)$$

3.2.4. Dynamic integration of users' long-short-term interests

When obtaining users' interests, the long-short-term interests of users may overlap with each other. The disentanglement method for long-short-term interests of users proposed in Section 3.2.3 can effectively avoid this situation, but it is still difficult to effectively integrate these two interests. Most of the existing methods aggregate these two interests through aggregation functions, and the importance of long-term and short-term interests on users' interests is set as fixed, which is not conducive to distinguishing and utilizing users' long-short-term interests. Consequently, we propose a dynamic integration method for users' long-term and short-term interests. This method inputs the historical interaction sequences into the GRU and learns the importance τ of users' long-term interests by attention mechanism. The importance of users' short-term interests is represented as $(1 - \tau)$. User u 's interest u_{int} at time t as formulated in Eqs. (23)–(25).

$$h_u^t = \text{GRU}[H_u^1, H_u^2, \dots, H_u^T] \quad (23)$$

$$\tau = \sigma(\text{MLP}[\leq_u^t \| L_u^t \| S_u^t]) \quad (24)$$

$$u_{int} = \tau \cdot L_u^t + (1 - \tau) S_u^t \quad (25)$$

where h_u^t represents the historical interaction of the user at time t . We use the MLP to link users' long-term interests and short-term interests. Then, calculate the interaction prediction score between the user u and the item v , as given in Eq. (26).

$$y_{int} = \text{MLP}[u_{int} \| e^v] \quad (26)$$

According to the literature (Yu et al., 2019), the negative logarithmic likelihood loss function is used to learn prediction scores, as given in Eq. (27).

$$\mathcal{L}_{pre} = -\frac{1}{N} \sum \hat{y}_{int} \log(y_{int}) + (1 - \hat{y}_{int}) \log(1 - y_{int}) \quad (27)$$

where N represents all training instances, $\hat{y}_{int} = 1$ represents that the user interacts with the item, and $\hat{y}_{int} = 0$ represents the opposite. Therefore, the joint loss function for the acquisition of users' interests stage as formulated in Eq. (28).

$$\mathcal{L}_u = \mathcal{L}_{int} + \mathcal{L}_{pre} + \lambda \|\Theta\|_2 \quad (28)$$

where $\lambda \|\Theta\|_2$ represents L2 regularization of Θ parameterization by λ to prevent overfitting.

The algorithm for the acquisition of users' interests is described as follows:

Algorithm Disentanglement and dynamic integration of users' long-short-term interests**Input:** user embedding e^u , item embedding e^v , where $e^u, e^v \in \mathbb{R}^k$, user history interaction behavior sequence $\mathcal{H}_u$ **Output:** user interest scoring y_{int}

- 1: **User long-term interest acquisition:**
- 2: $v_l \leftarrow \sigma(W_1 \cdot e^v)$
- 3: $\alpha_v \leftarrow MLP[v_l \| e^u \| (v_l \cdot e^u)]$
- 4: $L_u^l \leftarrow \sum_{v=1}^t \tilde{\alpha}_v \cdot e^v$, where $\tilde{\alpha}_v = softmax(\alpha_v)$
- 5: **User short-term interest acquisition:**
- 6: $[e_1^v, e_2^v, \dots, e_t^v] \leftarrow GRU[\mathcal{H}_u^1, \mathcal{H}_u^2, \dots, \mathcal{H}_u^T]$
- 7: $v_s \leftarrow \sigma(W_2 \cdot e_s^v)$
- 8: $\beta_v \leftarrow MLP[v_s \| e^u \| (v_s \cdot e^u)]$
- 9: $S_u^t \leftarrow \sum_{v=1}^t \tilde{\beta}_v \cdot e^v$, where $\tilde{\beta}_v = softmax(\beta_v)$
- 10: **User long-short-term interests disentanglement:**
- 11: $L_u^t \leftarrow \frac{1}{t} \sum_{v=1}^t e^v$
- 12: $S_u^t \leftarrow \frac{1}{k} \sum_{v=1}^k e^{t-v+1}$
- 13: $L_u^t, S_u^t \leftarrow ContrastLearning(L_u^t, S_u^t, L_u^t, S_u^t)$
- 14: **Dynamic integration of users' long-short-term interests:**
- 15: $h_u^t \leftarrow GRU[\mathcal{H}_u^1, \mathcal{H}_u^2, \dots, \mathcal{H}_u^T]$
- 16: $\tau \leftarrow \sigma(MLP[\|L_u^t\| \|S_u^t\|])$
- 17: $u_{int} \leftarrow \tau \cdot L_u^t + (1 - \tau) S_u^t$
- 18: $y_{int} \leftarrow MLP[u_{int} \| e^v]$

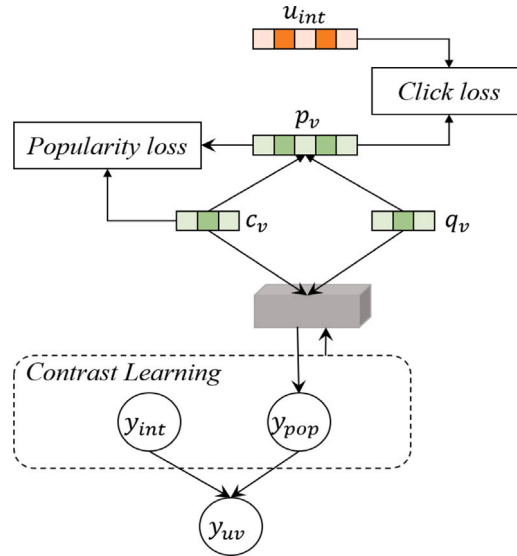


Fig. 5. The framework for disentanglement of user-item interactions.

3.3. Disentanglement of user-item interactions

In existing models (Zhang et al., 2021; Zheng et al., 2021) that disentangle user interest and popularity, item popularity is usually expressed through conformity, neglecting the influence of attributes (i.e., item quality) on item popularity. In this paper, we consider that there are two factors affecting item popularity which are item attributes and conformity. The disentanglement of the user-item interactions framework is demonstrated in Fig. 5.

In Section 3.2, we disentangled users' long-short-term interests by contrastive tasks, and integrate them dynamically to improve the controllability and interpretability of the model. Similarly, according to formula (1), we design the following contrastive learning tasks to disentangle the reasons for user-item interaction. We divided the training instances into ω_1 and ω_2 . For positive item i and negative item j , user interaction with the item can be divided into the following two cases:

- (1) In ω_1 , positive item i is more popular than negative item j , and the user selects the popular item i to interact instead of the unpopular negative item j , then there is the inequality as given in Eqs. (29) and (30).

$$y_{ui}^{pop} > y_{uj}^{pop} \quad (29)$$

$$y_{ui}^{int} + y_{ui}^{pop} > y_{uj}^{int} + y_{uj}^{pop} \quad (30)$$

- (2) In ω_2 , negative item j is more popular than positive item i , and the user chooses to interact with the unpopular item i instead of popular negative item j , then there is the inequality as given in Eqs. (31)–(33).

$$y_{ui}^{pop} < y_{uj}^{pop} \quad (31)$$

$$y_{ui}^{int} > y_{uj}^{int} \quad (32)$$

$$y_{ui}^{int} + y_{ui}^{pop} > y_{uj}^{int} + y_{uj}^{pop} \quad (33)$$

Through the above contrastive learning, we can achieve the disentanglement of the two reasons for user–item interaction. Based on the above two cases, we use different training instances to optimize the popularity embedding, and use the BPR loss function to model the item popularity, as formulated in Eq. (34).

$$\mathcal{L}_{pop} = \sum_{(u,i,j) \in \omega_1} BPR(\langle p_v, c_i \rangle, \langle p_v, c_j \rangle) + \sum_{(u,i,j) \in \omega_2} -BPR(\langle p_v, c_i \rangle, \langle p_v, c_j \rangle) \quad (34)$$

Finally, according to formula (1), we combine the two reasons that affect user–item interaction and recommend items with the top k of the user–item interaction score to the user. Using the BPR loss function to make the recommendation score of positive item i higher than negative item j , as formulated in Eq. (35).

$$\mathcal{L}_{click} = \sum_{(u,i,j) \in N} BPR(\langle u_{int}, p_i \rangle, \langle u_{int}, p_j \rangle) \quad (35)$$

In the DDCE model, we design different modeling mechanisms and loss functions for different tasks. To learn these tasks simultaneously, we combine these loss functions, as given in Eq. (36).

$$\mathcal{L} = \mathcal{L}_{click} + \mathcal{L}_u + \mathcal{L}_{pop} \quad (36)$$

The algorithm for the disentanglement of user–item interactions is described as follows:

Algorithm Disentanglement of user–item interactions

Input: item popularity scoring y_{pop} , item popularity p_v , conformity c_v , user interest u_{int} , user interest scoring y_{int} , joint loss function for the acquisition of users' interests stage $\mathcal{L}_u$

Output: user–item interaction scoring y_{uv}

- 1: $y_{int}, y_{pop} \leftarrow ContrastLearning(y_{ui}^{int}, y_{uj}^{int}, y_{ui}^{pop}, y_{uj}^{pop})$
 - 2: $\mathcal{L}_{pop} \leftarrow \sum_{(u,i,j) \in \omega_1} BPR(\langle p_v, c_i \rangle, \langle p_v, c_j \rangle) + \sum_{(u,i,j) \in \omega_2} -BPR(\langle p_v, c_i \rangle, \langle p_v, c_j \rangle)$
 - 3: $\mathcal{L}_{click} \leftarrow \sum_{(u,i,j) \in N} BPR(\langle u_{int}, p_i \rangle, \langle u_{int}, p_j \rangle)$
 - 4: $\mathcal{L} \leftarrow \mathcal{L}_{click} + \mathcal{L}_u + \mathcal{L}_{pop}$
 - 5: $y_{uv} \leftarrow y_{int} + y_{pop}$
-

4. Experiment

In this section, we compare the proposed model DDCE with benchmark models to verify its effectiveness. To provide a more comprehensive evaluation of the proposed model, our experiment aims to answer the following five research questions:

- RQ1: Is DDCE better than the SOTA method? (Section 4.2)
 RQ2: How to verify the effectiveness of dynamic integration of user long-term and short-term interests in the model? (Section 4.3.1)
 RQ3: Does DDCE disentangle users' long-term and short-term interests? (Section 4.3.2)
 RQ4: What is the influence of item attributes and conformity on recommendation performance? (Section 4.3.3)
 RQ5: What are the influences of user interest and item popularity on the model? Does DDCE disentangle user–item interactions? (Section 4.3.4)

4.1. Experiment settings

This subsection describes the datasets used in the experiment, benchmark models, and the experimental implementation details.

Table 3
Statistics of datasets.

	# users	# items	# user–item interactions	# Interaction density
Douban Movie	45,923	24,658	6,717,919	0.5932%
KuaiRec	7,176	10,362	12,528,859	16.8494%

4.1.1. Datasets

To evaluate the proposed model DDCE, we conduct experiments on two real datasets, Douban Movie and KuaiRec, which are both publicly available and collected from real applications and differ in sparsity and field. The statistics of the dataset are shown in Table 3 :

Douban Movie: On Douban, users can comment the books they have read, the movies they have watched, and the music they have listened to. In this paper, we use the Douban movie dataset¹ collected by literature (Song et al., 2019), which contains users' comments on movies and relevant timestamps. Each rating in the dataset represents the interaction between the user and the movie. To ensure sufficient feedback from each user or item, we conduct 10-core filtering of the dataset, which means filtering out users and items with less than 10 interaction records.

KuaiRec (Gao, Li, et al., 2022): This dataset² is collected from the recommendation logs of the Kwai App, a short video platform in the real world. On this platform, users can watch various types of short videos, and users can switch the video they are currently watching at any time without waiting for the end of the video playback. The dataset contains the interaction records between users and videos, as well as relevant timestamp information. We filter out items with less than 10 interactions in the dataset.

4.1.2. Evaluation metrics

To evaluate the accurate performance of model recommendation, this paper uses three frequently-used evaluation indicators for recommender systems, which are *NDCG* (normalized discount cumulative gain), *HR* (click-through rate), and *Recall* (recall rate). The higher the value of these three indicators, the better the model's accurate performance.

- (1) *NDCG@K*: focus on the hit position of the item. If the hit item is in front, it will receive a higher score, the formula as given in Eq. (37).

$$NDCG@K = \frac{1}{|U|} \sum_{u \in U} \frac{\sum_{a=1}^K 2^{\delta_u(a)} - 1}{\log_2(a+1)} \quad (37)$$

where $\delta_u(a) \in \{0, 1\}$ is a binary indicator function, $\delta_u(a)$ returns 1 if the recommendation at position a receives feedback, otherwise the value is 0.

- (2) *HR@K*: is used to verify whether the recommendation sequence contains items that the user interacts with, the formula as given in Eq. (38).

$$HR@K = \frac{1}{|U|} \sum_{u \in U} \frac{\sum_{a \leq K} \delta_u(a)}{K} \quad (38)$$

- (3) *Recall@K*: is used to measure how many positive samples are predicted to be positive, the formula as given in Eq. (39).

$$Recall@K = \frac{1}{|U|} \sum_{u \in U} \frac{\sum_{a \leq K} \delta_u(a)}{\sum_a \delta_u(a)} \quad (39)$$

- (4) *Novelty@K*: In recommender systems, popularity bias often results in less novelty. Therefore, we measure the model's ability to recommend novel items to users by novelty (Zhou et al., 2010). For item i , the calculation of novelty (Zhao, Wu, et al., 2022) is shown in Eq. (40).

$$Novelty@K = \frac{1}{|U| K} \sum_{u \in U} \sum_{i \in I_u(K)} -\frac{1}{\log_2 |U|} \log_2 \left(\frac{d_i}{|U|} \right) \quad (40)$$

where $I_u(K)$ represents the top- K recommendation list of user u in the test data, and d_i represents the number of interactions observed in the training data for item i .

4.1.3. Baselines

To evaluate the performance of the DDCE model proposed in this paper, we compare it with the following SOTA methods that solve the popularity bias problem in recommendations by using causal reasoning:

- IPS (Schnabel et al., 2016) is the inverse propensity score model. The instance weight is set to the reciprocal of the popularity value of the corresponding item, so that popular items are assigned a lower weight, and increasing the weight of long-tail items is to eliminate the popularity bias.

¹ <https://github.com/DeepGraphLearning/RecommenderSystems/blob/master/socialRec/README.md#douban-data>

² <https://kuaiRec.com>

Table 4

Overall accurate performance comparison of all methods on Douban Movie and KuaiRec datasets for $K = 20$. The improvement achieved by DDCE is significant (p -value < 0.05).

Backbone	Method	Douban Movie			KuaiRec		
		NDCG	HR	Recall	NDCG	HR	Recall
MF	–	0.0442	0.1791	0.0269	0.0058	0.0282	0.0072
	IPS	0.0440	0.1790	0.0266	0.0058	0.0280	0.0071
	CausE	0.0439	0.1788	0.0264	0.0051	0.0277	0.0067
	DICE	0.0462	0.1819	0.0291	0.0062	0.0379	0.0089
	MACR	0.0484	0.2217	0.0312	0.0078	0.0416	0.0103
	PDA	<u>0.0683</u>	<u>0.2893</u>	<u>0.0408</u>	<u>0.0099</u>	<u>0.0475</u>	<u>0.0135</u>
	DDCE w/o i	0.0591	0.2802	0.0386	0.0081	0.0428	0.0115
	DDCE	0.0717	0.2994	0.0436	0.0103	0.0490	0.0142
	Improv.	4.9780%	3.4911%	6.8627%	4.0404%	3.1578%	5.1851%
LightGCN	–	0.0508	0.2761	0.1286	0.0074	0.0303	0.0163
	IPS	0.0503	0.2648	0.1253	0.0065	0.0284	0.0158
	CausE	0.0516	0.2789	0.1307	0.0076	0.0328	0.0172
	DICE	0.0535	0.2808	0.1369	0.0089	0.0393	0.0205
	MACR	0.0578	0.2977	0.1429	0.0101	0.0460	0.0231
	PDA	<u>0.0784</u>	<u>0.3016</u>	<u>0.1446</u>	<u>0.0118</u>	<u>0.0498</u>	<u>0.0248</u>
	DDCE w/o i	0.0677	0.3002	0.1438	0.0107	0.0485	0.0242
	DDCE	0.0812	0.3098	0.1521	0.0122	0.0507	0.0259
	Improv.	3.5714%	2.7188%	5.1867%	3.3898%	1.8072%	4.4354%

- CausE (Bonner & Vasile, 2018) sets up two sets of embeddings for each user and item on biased data and unbiased data, respectively. And applies regularization to make these embeddings similar.
- DICE (Zheng et al., 2021) uses causal reasoning to disentangle user interest and conformity, which eliminates popularity bias from the user viewpoint.
- MACR (Wei et al., 2021) uses the causal graph to train the recommendation model. This method evaluates the importance of different reasons for ranking scores and conducts counterfactual reasoning to eliminate popularity bias during the recommendation stage.
- PDA (Zhang et al., 2021) uses the causal intervention to set the recommendation score to the expected. This method analyzes the effect of item popularity on recommendation from the causal perspective.

4.1.4. Experimental implementation details

After the literature (Zheng et al., 2021), the causal recommendation is often used as an additional method to the backbone recommendation model, so we use the commonly used recommendation model matrix decomposition (MF) and graph-based model LightGCN as the backbone model to compare different methods.

In this paper, we use Python 3.6.7 and PyTorch for experiments. We first ordered user–item interactions by timestamp. For each data set, we divide the training set/validation set/test set into 8/1/1. That is, 80% of the data is used to learn the parameters, 10% is used to optimize the hyperparameters, and the remaining data is used for final performance comparison. Because the other models except the IPS model contain two sets of embeddings, to make the experimental results fair, we fixed the embedding size of the IPS as 128, and the embedding size of the other models as 64. The hyperparameters are estimated using the Adam (Kingma & Ba, 2014) optimizer during model learning. To obtain the optimal parameters of the model, we use the grid search algorithm. The batch size is 1024, the regularization coefficient of L2 is $1e-5$, and the learning rate is $\eta = 0.001$. For the parameter settings of the benchmark model, we use the parameters in the original article. In the meantime, to guarantee the optimal performance of the benchmark model, a grid search is used to fine-tune the hyperparameters. To avoid overfitting problems, we implement an early stop strategy to cease the training when the performance declines for ten consecutive epochs on the verification set.

4.2. Experimental results and analysis (RQ1)

In this subsection, we first conduct experimental research and analysis on the accurate performance of the model, and then conduct experimental research on the novelty of the model.

4.2.1. Model accuracy research

We compare the proposed model DDCE with the benchmark model and the basic MF and LightGCN models (i.e. without any causal methods) on the Douban Movie dataset and the KuaiRec dataset. For each model, we consider the highest result at $K = 20$ as the best result. We summarize the experimental results of all models in Table 4 and then analyze the results. The representation of the optimal result is bold, and the representation of the suboptimal result is underlined.

We analyze the experimental results from Table 4, and summarize the findings as follows:

For causal embedding recommendation models based on MF and LightGCN, our model achieved the best performance in NDCG, HR, and Recall on two different types of real datasets, Douban Movie and KuaiRec. This indicates that the dual disentanglement of user long–short-term interests, user interest and item popularity have a positive effect on recommendation results.

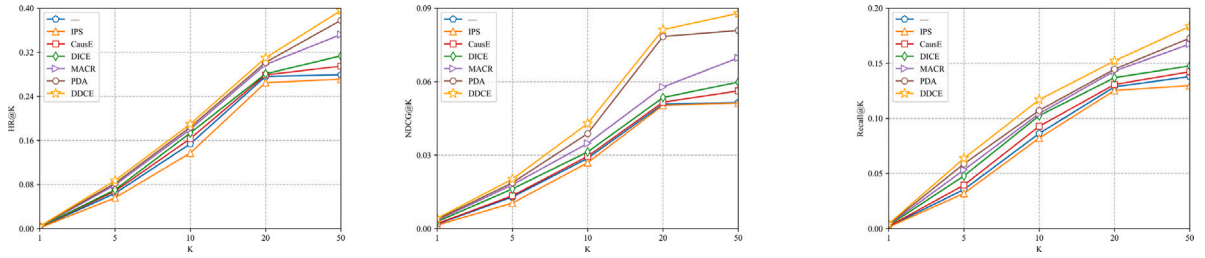


Fig. 6. Top-K recommendation performance on the Douban Movie dataset with LightGCN as the backbone.

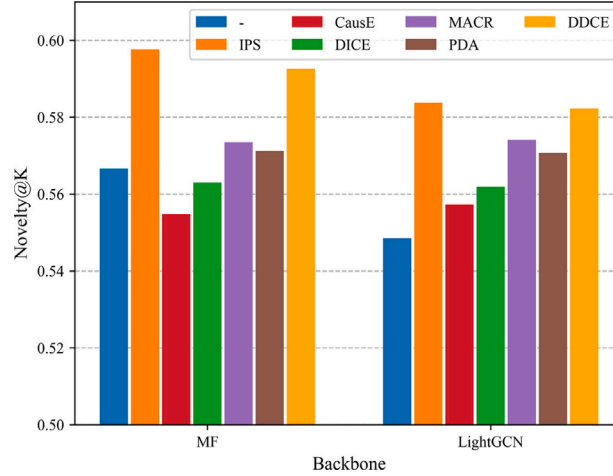


Fig. 7. Recommendation novelty of methods on the Douban Movie dataset.

Among these two backbones, the results of DICE are better than IPS and CausE, indicating that disentanglement of user interest and conformity can increase the recommended performance effectively. The performance of MACR and PDA is better than that of DICE, because in the original DICE paper, to ensure a uniform distribution of interactions across items, manual intervention is performed on training data, which eliminated the popularity bias in test data. In our experiment, user-item interactions are sorted by timestamp for each dataset to reflect the recommended performance of the model in real-world recommender system applications. MACR and PDA have analyzed and used conformity in the model, thus achieving better results than DICE. PDA adjusts the popularity bias during the inference stage, so the recommendation performance is better than MACR. Our model DDCE first disentangles users' long-short-term interests and then disentangles user interest and item popularity. When obtaining item popularity, the influence of item attributes is considered, so that the best recommendation performance is achieved.

DDCE w/o i in the table represents that the disentanglement of user-item interaction neglects the disentanglement of users' long-short-term interests, and other parts of the model remain unchanged. From the experimental results in the table, the performance of DDCE w/o i is better than DICE and MACR, but slightly inferior to PDA, indicating that disentanglement of users' long-short-term interests can effectively improve the model quality. The dual disentanglement of user interests and user-item interaction through contrastive learning tasks enables DDCE to obtain more accurate recommendations in real-world recommender system applications while ensuring robustness and improving the interpretability of the model.

From Table 4, we find that the recommendation effect of LightGCN as the backbone is better than MF. Therefore, we use the LightGCN as the backbone to study the effect of the K value (where $K \in \{1, 5, 10, 20, 50\}$) on recommendation performance. The experimental results of different K values on the Douban Movie dataset are demonstrated in Fig. 6. According to the figure, we find that at different K values, our model DDCE outperforms all baselines, representing its robustness.

4.2.2. Model novelty research

Popularity bias usually leads to low novelty in popular items. To verify the effectiveness of the model in dealing with the problem of popularity bias, we studied the novelty of the model on the dataset of Douban Movie. The experimental results are shown in Fig. 7, where $K = 20$.

From Fig. 7, we find that when MF and LightGCN are used as backbone models, the novelty recommended by IPS is the best, while the other methods are inferior to IPS in terms of novelty. This is because the IPS reweights each instance by the popularity of the item to reduce the popularity bias at the cost of reducing the accuracy of the recommendation. The weight of the instance is set

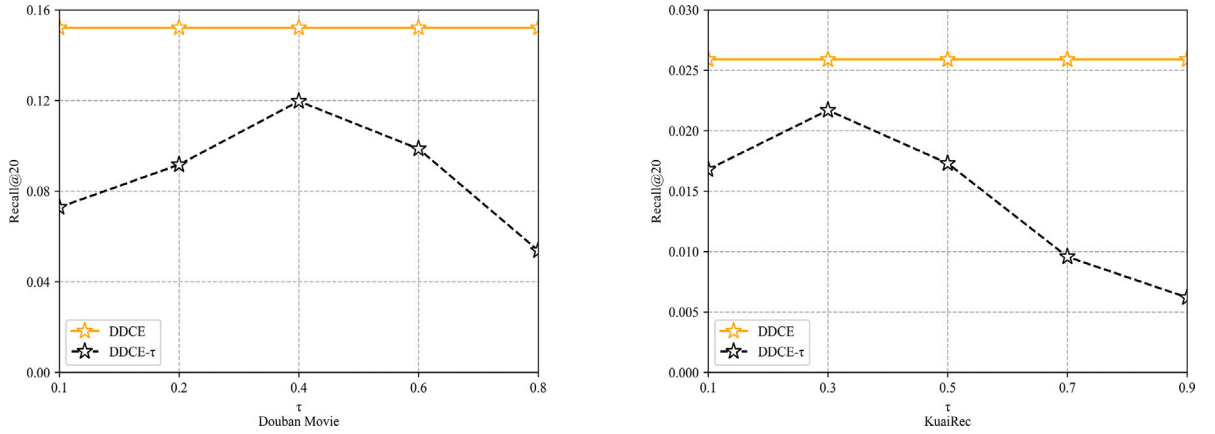


Fig. 8. The analysis of dynamic integration users' long-short-term interests.

to the reciprocal of the corresponding item popularity value, so the weight of the item in the long tail is increased. Novelty is usually negatively correlated with the popularity of the item, so the item recommended by IPS to the user has a higher novelty. Although other benchmark causal recommendation methods have shown some effectiveness in dealing with the problem of popularity bias, the method proposed in this paper has better performance in improving the novelty of recommendations.

4.3. Model learning

In this subsection, we conducted ablation experiments to future research the influence of various parts of DDCE on recommended performance. According to the experimental results in Section 4.2, we found that the recommended effect of using LightGCN as the backbone is better than MF. Therefore, we use the LightGCN as the backbone in the ablation experiments.

4.3.1. The influence of dynamic integration users' long- and short-term interests (RQ2)

In DDCE, dynamically integrating long-short-term interests is acquired through the attention mechanism. To verify whether dynamic integration is effective, after disentanglement of users' long-short-term interests, we set variant DDCE- τ , and use fixed τ value to integrate users' long-short-term interests. The disentanglement of user-item interaction remains unchanged. Fig. 8 demonstrates the recommended performance of DDCE and DDCE- τ on Douban Movie and KuaiRec datasets, where $K = 20$, the dynamic integration result represents by the solid line, and the fixed value integration result represents by the dotted line.

From Fig. 8, we find using the attention mechanism to dynamic integration users' long-short-term interests on Douban Movie and KuaiRec datasets can be better than different fixed τ values, which indicates the effectiveness of our dynamic integration method. On the Douban Movie dataset, there is a small difference in the impact of users' long-term interests and short-term interests on recommendations. On the KuaiRec dataset, we find that recommendation performance gradually decreases with the growth of τ , which indicate short-term interests of users on this dataset have a greater impact on the interaction between users and items. The different results on the two datasets indicate that the dynamic integration of users' long-short-term interests proposed in this paper can better adapt to different data types in real recommender systems.

There are experimental studies (Yu et al., 2019; Zheng et al., 2022) on the effect of users' long-term interest, short-term interest, and integration of long-short-term interests on the recommended performance. In this study, the contribution is that when disentanglement user-item interaction, users' long-short-term interests are first disentanglement through the contrastive learning task, and then dynamic integration is performed using the attention mechanism. Therefore, this paper does not conduct experiments on the influence of users' unilateral interests on recommendation performance.

4.3.2. The analysis of disentanglement users' long-short-term interests (RQ3)

To study whether DDCE disentangles users' long-short-term interests, we conducted a counterfactual assessment of the changes in the importance of different interests. Using the user's click and like behavior on the KuaiRec dataset, the user's like behavior is more dependent on the user's long-term interests, while the click behavior reflects the user's short-term interests. In this paper, the interactive behavior of users in the KuaiRec dataset where their video-watching length is more than twice that of the video itself is considered as user's like behavior. The importance of long-term and short-term interests is different when predicting users' click and like behavior. Because users' like behavior reflect their long-term interests, therefore, the importance of long-term interest τ should be more important than click behavior (short-term interest) $1-\tau$ when predicting like behavior. We compare the HR value and the average of τ for DDCE w/o i (i.e., without disentangling of users' long-short-term interests) with DDCE. Table 5 summarizes the experimental results.

Table 5
Performance comparison between DDCE w/o i and DDCE on the KuaiRec dataset.

Backbone	Method	Click		Like	
		HR	AVG τ	HR	AVG τ
LightGCN	DDCE w/o i	0.0485	0.7154	0.0447	0.7303
	DDCE	0.0507	0.2628	0.0461	0.2810

Table 6
Performance comparison of DDCE- q_v , DDCE- c_v , and DDCE.

Backbone	Method	Douban Movie			KuaiRec		
		NDCG	HR	Recall	NDCG	HR	Recall
LightGCN	DDCE- q_v	0.0788	0.3027	0.1459	0.0115	0.0497	0.0249
	DDCE- c_v	0.0793	0.3034	0.1477	0.0117	0.0501	0.0253
	DDCE	0.0812	0.3098	0.1521	0.0122	0.0507	0.0259

We analyze the results in Table 5 and find that the performance of DDCE is better than that of DDCE w/o i in both user interactions, and the average value of τ is much lower. The low average value of τ in DDCE indicates that short-term interests are more important than long-term interests, which is consistent with the conclusion in Section 4.3.1 that recommendations on the KuaiRec dataset are more dependent on users' short-term interests. This represents that our method can achieve the disentanglement of users' long-short-term interests in user-item interaction, and effectively improve the accuracy of recommendations. The high average value of τ in DDCE w/o i indicates that users' long-term interests learned by the model include more short-term interests, which are entangled with each other.

4.3.3. The influence of item attributes and conformity (RQ4)

In the DDCE model of this paper, the item popularity consists of item attributes and conformity, rather than being decided by single conformity. To better understand the influence of item attributes q_v and conformity c_v on recommendation performance, we set the following two variables:

- DDCE- q_v : represents that the item popularity is only determined by the attribute of the item, without considering the conformity of the item.
- DDCE- c_v : represents that the item popularity is directly expressed by conformity, without considering the item attribute.

We compare these two variables with the complete DDCE and summarize the experimental results in Table 6, where $K = 20$.

From the experimental results in Table 6, the performance of DDCE is the best on both datasets and DDCE- c_v is better than DDCE- q_v . For item popularity, both item attributes and conformity influence it. The influence of attributes is slightly lower than conformity, which indicates that when users interact with the item, conformity drives users more than the attribute of the item. Our experimental results indicate that the combination of the two can make the item popularity more complete and have a positive effect on improving recommendation performance.

4.3.4. The analysis of disentanglement user-item interaction (RQ5)

According to Fig. 3 cause graph, we consider that the interaction between the user and the item depends on the matching of user interests and item popularity. To research the influence of user interest and item popularity on the model, we set up two variants, DDCE- int and DDCE- pop .

- DDCE- int : indicates that only user interests are used in recommendations.
- DDCE- pop : indicates that only item popularity is considered when making recommendations to users.

Fig. 9 shows the experimental results of DDCE- int , DDCE- pop , and DDCE on the KuaiRec Dataset.

To analyze the performance comparison results in Fig. 9, we find that the DDCE has the best performance compared with DDCE- int and DDCE- pop , and the results of DDCE- pop are slightly worse than DDCE- int . This indicates that user interest has a greater influence on interaction than item popularity, and the combination of the two can effectively improve overall performance. To verify whether DDCE disentangles user-item interaction, we compare the overlapping recommended items of DDCE, DDCE- int , and DDCE- pop with ItemPop (Rendle et al., 2012). In ItemPop, items are ranked according to popularity and interaction times, and recommend the most popular item. We use intersection over union (IOU) as the evaluation indicator, and the higher the value of the IOU, the more overlapping items are recommended. We use Intersection Over Union (IOU) as the evaluation indicator, and the higher the value of the IOU, the more overlapping items are recommended. Fig. 10 shows the results of disentanglement user-item interaction on the KuaiRec dataset.

From Fig. 10, we observe that DDCE- int has little interaction with popular items, indicating that popularity information is almost completely disentangled from interest embedding. With the increase of K value, the overlapping items in DDCE- pop gradually increased, and the value of DDCE- pop is always higher than DDCE, and DDCE is relatively stable. It indicates that our method can

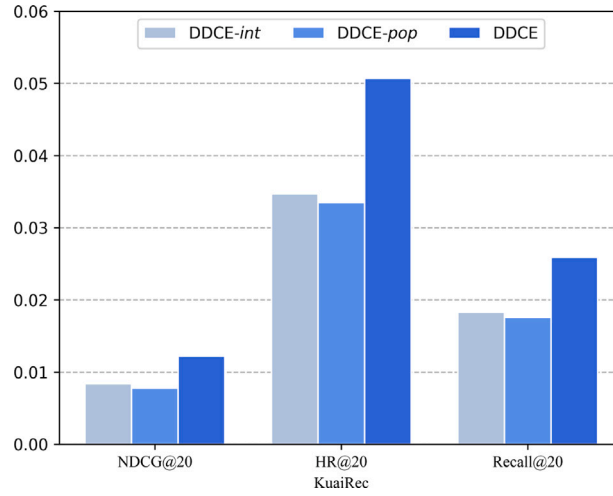


Fig. 9. Performance comparison of DDCE-int, DDCE-pop, and DDCE on the KuaiRec Dataset.

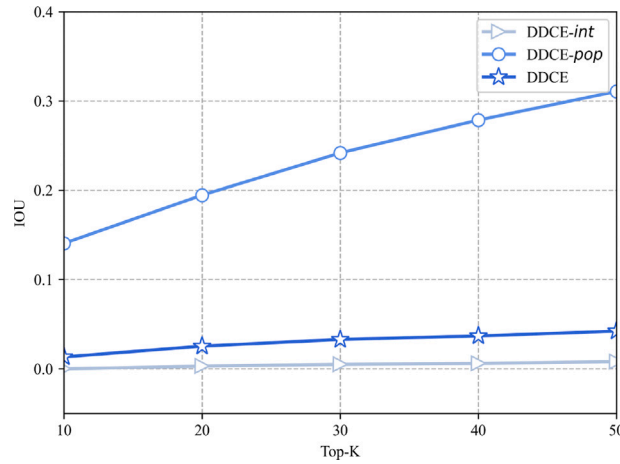


Fig. 10. The analysis of disentanglement user-item interaction on the KuaiRec dataset.

achieve the disentanglement of user interest and item popularity. Through the analysis of user-item interaction disentanglement, we have the following conclusions: user interest and item popularity have different influences on recommendation, and the combination of the two can effectively improve recommendation performance. Disentanglement can effectively prevent the recommender system from recommending too many popular items to users, which makes the user bored with the recommended content and reduce the user's satisfaction with the recommendation result.

5. Conclusion and future work

To address the issue of popularity bias research based on causal reasoning in recommender systems, we propose a novel dual disentanglement of user-item interaction for recommendation with causal embedding. The model firstly analyzes the causality of user-item interaction and obtains the cause embedding of each part. Then, the popularity bias is studied on the user-side and item-side, respectively. On the item-side, the model considers the item attributes on popularity to enhance the dependability of the item's popularity. On the user-side, we divide user interests into long-term interests and short-term interests and disentangle and dynamically integrate them to be better applied to different recommendation scenarios. Finally, the causes for user-item interaction are disentangled by the disentanglement of user interest and item popularity, which enhances the stability and interpretability of the model. We verify the effectiveness of DDCE on Douban Movie and KuaiRec datasets. In the ablation experiment, we analyze the dual disentanglement to prove that DDCE can achieve the disentanglement of users' long-short-term interests and the disentanglement of user-item interaction.

The theoretical and practical implications of our research are that we consider the influence of popularity bias on recommendation from a double-end perspective when studying popularity bias based on causal embedding, which makes the study of popularity

bias more comprehensive. We consider item attributes in the research of item popularity which is helpful to alleviate the problem of item-side popularity bias. We research the disentanglement and dynamic integration of users' long-short-term interests, which makes our recommendation model closer to the real-world recommender system. In the model proposed in this paper, we did not consider the user's social relationship, and the user's interest will be affected by friends. Therefore, our future research will consider users' social relationships when studying users' interests.

CRedit authorship contribution statement

Chenyu Wang: Conceptualization, Methodology, Validation, Formal analysis, Writing – original draft, Project administration. **Yawen Ye:** Investigation, Validation. **Liyuan Ma:** Investigation, Data curation, Software. **Dun Li:** Resources, Writing – review & editing, Supervision. **Lei Zhuang:** Resources, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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